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Fashion Recommender System Design

1. Scope

Domain and Intended Users of the Recommender System

The domain of the recommender system is fashion retail, specifically tailored for H&M customers. This system aims to provide personalized recommendations for clothing, shoes, accessories, and jewellery. There are 132 unique product types in the H&M dataset. It focuses on enhancing the shopping experience by offering suggestions that match individual preferences and current fashion trends.

The intended users of this system are diverse, including regular shoppers, fashion enthusiasts, occasional shoppers, and new users. Regular shoppers frequently visit H&M, either online or in-store, looking for new arrivals and promotions. Fashion enthusiasts are deeply engaged with the latest trends and often seek out the newest fashion items. Occasional shoppers might visit H&M for specific events like marriage etc. or seasonal wardrobe updates and rely on recommendations to discover new products. New users, or first-time visitors, depend heavily on initial recommendations and popular and highly rated items to guide their shopping.

The user interaction will be facilitated through both web and mobile interfaces. The web interface allows for a more extensive display due to the larger screen size, which can accommodate more items without overwhelming the user. The web interface will display up to 20 items, providing detailed information about each product, including images, prices, ratings, and reviews. The mobile interface, optimized for smaller screens, will show up to 10 items at a time. Users can interact with the recommendations by liking, disliking, or rating the items, helping to refine future suggestions. For the purpose of this project design, the number of recommended items is chosen as 15, which is a balance between mobile and web interfaces.



MOBILE INTERFACE



WEB INTERFACE

To keep the recommendations relevant and personalized, the system will dynamically update based on user interactions and feedback. This includes adjusting the weighting of different recommendation methods (content-based, collaborative filtering, etc.) to improve accuracy. i.e. it will employ hybrid ensemble method. Regarding whether this is feasible, we only plan to use a small subset of the H&M dataset for this project so as to train the model and test it. Hence, real time updating won't be a time consuming or computationally expensive task.

To address cold start problems, the system will initially recommend popular and trending items that have received positive feedback from other users. New users will be prompted to complete a quick survey or questionnaire about their style preferences, favourite categories, and sizes. Additionally, demographic information such as age, gender, and location will be used to tailor recommendations further.

The business model of the recommender system aims to increase user engagement and drive sales through personalized suggestions. This will be achieved by providing relevant recommendations that encourage more frequent purchases and higher order values. The system will also support cross-selling and up-selling by suggesting complementary items and higher-end alternatives. Furthermore, premium features like early access to new collections and personalized styling advice can be offered as part of a subscription service. The recommendation feed may also include targeted advertisements to generate additional revenue.

In summary, this recommender system is designed to cater to a wide range of users, offering a seamless and engaging shopping experience. By leveraging user feedback and demographic information, it aims to provide personalized fashion recommendations that enhance user satisfaction and drive business growth for H&M.

2. Dataset

I have selected H&M Personalized Fashion Recommendations dataset for this project. The source of this dataset is this link from Kaggle: [All-in-One Sample Dataset \(kaggle.com\)](https://www.kaggle.com/datasets/hugoboss/hm-personalized-fashion-recommendations)

There is an extra images folder available in Kaggle. It has the image of each fashion item. Hence, for a full-fledged project, the recommendation model can show the image of the items that are recommended, in the visual interface like mobile . However, I am not implementing that for this project.

There are three ZIP files of different sizes: Small (5% of original dataset), Mini (1% of original dataset) and Tiny (0.2% of original dataset). I plan to create a combined dataset from all three.

First of all, there is no solution code for this dataset in the Kaggle website. The H&M dataset was used for some other coding challenge competition, but the coding challenge was just doing Exploratory Data Analysis. I have searched through Kaggle and no one has uploaded any code for creating a recommender system in Kaggle or any other site.

Each ZIP files has three datasets in them. They are dataset for articles, transactions and customers.

I have combined them all to a single dataset each for articles, customers and transactions.

The column or feature names of the articles dataset are article_id, product_code, prod_name, product_type_no, product_type_name, product_group_name, graphical_appearance_no, graphical_appearance_name, colour_group_code, colour_group_name, perceived_colour_value_id,

perceived_colour_value_name, perceived_colour_master_id, perceived_colour_master_name, department_no, department_name, index_code, index_name, index_group_no, index_group_name, section_no, section_name, garment_group_no, garment_group_name, detail_desc

The column or feature names of the customers dataset are customer_id, FN, Active, club_member_status, fashion_news_frequency, age, postal_code

The column or feature names of the transactions dataset are transaction_date, customer_id, article_id, price, sales_channel_id, month, year

Regarding scale and diversity of the dataset, there are 132 unique product types, there are 105543 unique fashion items. This is a very diverse and big dataset of fashion items of H&M.

There is data of 83,533 customers along with information of their ages, whether they have purchased atleast once in the site, and whether they are active customers and whether they hold premium club membership. This is also a big and diverse dataset, From the ages of customers, we can infer what age group purchases the most items, and target specific items for each age group, and increasing the chances that they buy a lot, thus increasing the business of H&M. We can also create the recommendation model such that it targets the “active but not club membership” and “active and club membership” customers separately. There is a chance that each of these two types of members may be of different financial status, and hence we can aim targeted ads and targeted recommended items for each. The aim is to make the “active but not club membership” customers to become a “active and club membership” customer. Similarly, the recommendation model must be good enough to convert a non-active customer or a rarely site-visiting customer a regular one.

However, for the purposes of this project, we can filter out the non-active customers to get a smaller subset of dataset to build a starter recommender model, which we can build later on to be a full-fledged recommender model.

A notable limitation of the H&M dataset is the absence of explicit ratings for items. Ratings are an important source of information for recommendation systems, as they provide direct feedback from users about their preferences. Without ratings, the system must rely more heavily on implicit feedback, such as purchase history, browsing behavior, and interaction data. That is what I plan to do for this H&M dataset. I will calculate purchase frequency and recency to infer user preferences. Then, by analyzing patterns in transactions of each user, and leveraging temporal trends of group of users, I will use these features in collaborative filtering techniques along with machine learning algorithms like LightGBM for ranking or rating items. This approach will help me circumvent lack of explicit ratings and help in building a robust recommender system.

There are no user reviews or remarks in the dataset. Sentiment Analysis is not always practical. Whether a user’s review or remark is positive or negative is very hard to decipher. Hence, the lack of user reviews in the dataset is not a big deal.

The dataset is huge. However, I will remove rows with lot of null fields. Then, I will only take a subset of the dataset. I can use age, active_status and membership_status etc. to filter out regular, active, and membership customers. Thereby, lot of inactive customers data will be removed, and the subset left will be smaller and give us more precise recommendations.

However, the good thing is since the original dataset itself is big, even after preprocessing and removal of redundant or un-useful data, there will still be a significant volume of data to train and validate the recommendation models. This large quantity of data will be useful in capturing diverse user behaviours

and preferences, which is crucial for developing a model that can generalize well across different user segments.

Regarding the temporality and the volume of the transactions data, the dataset is large, covering transactions from September 2018 to January 2020, and there are 1,048,576 transactions. Thus, this data is enough for the analysis of seasonal or annual trends and changes in user preferences over time.

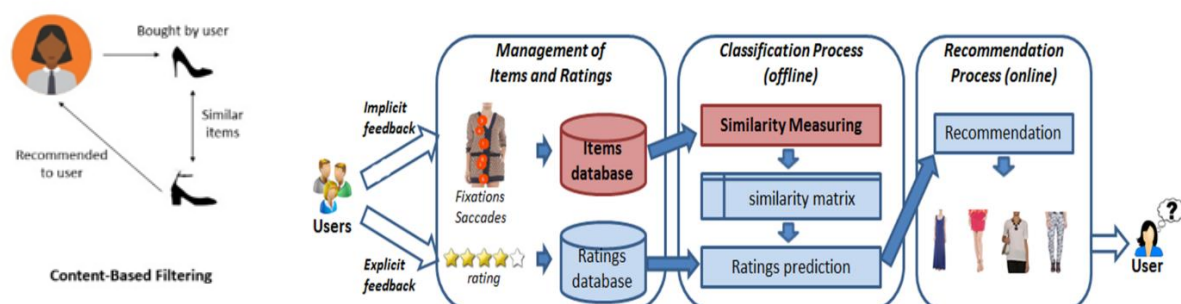
This dataset is realistic and comprehensive, including metadata about customers, articles, and transactions.

3. Methods

The Project spans for 6 weeks i.e. Week 6 to Week 11

Week 1: Content-Based Filtering

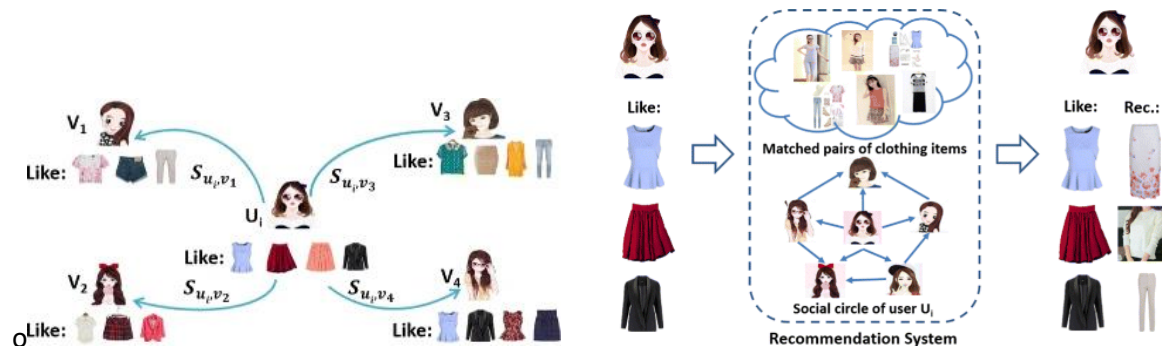
In the first week, we will focus on implementing a content-based filtering method. This approach leverages the metadata associated with the items in the dataset, such as product descriptions, categories, and attributes like color, product type, style, and brand. Content-based filtering is particularly suitable for the fashion domain because it can provide personalized recommendations based on the specific features of the items that a user has previously interacted with. For instance, if a user frequently purchases black dresses, the system can recommend other dresses in similar styles or colors. This method is essential for new users who lack historical interaction data, as it can make initial recommendations based on the explicit preferences they provide. Thus, it can resolve cold start problem. We can imagine this in real world scenario, as for the first 3 days after a user has started visiting the website for the first time, only content based filtering would be used to recommend popular and trending items that have received positive feedback from other users. New users will be prompted to complete a quick survey or questionnaire about their style preferences, favourite categories, and sizes. Additionally, demographic information such as age, gender, and location will be used to tailor recommendations further. We will use methods like TF-IDF Vectorization and then Cosine Similarity to get recommendations of similar items.



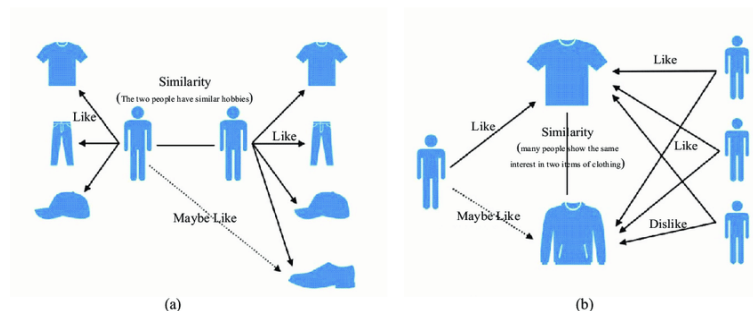
Week 2: Collaborative Filtering

In the second week, I will develop a collaborative filtering model using implicit feedback derived from the transaction data. This method is effective because it identifies patterns in user behavior to recommend items that similar users have liked. Since the H&M dataset does not contain explicit ratings, we will infer preferences from metrics such as purchase frequency and recency. Collaborative filtering is well-suited for active users with sufficient interaction history, as it can uncover latent

preferences and suggest items that users with similar behavior have enjoyed. Since I am only filtering out subset of active users from original dataset, the data will have less noise. This approach also benefits from the large volume of data, making the recommendations more accurate and diverse. Some of the methods I can use for Collaborative Filtering are k-Nearest Neighbours and Matrix factorization.



Similarity matrix of users, items will be used as shown in the above figure.



Two types of user similarity matrices can be used

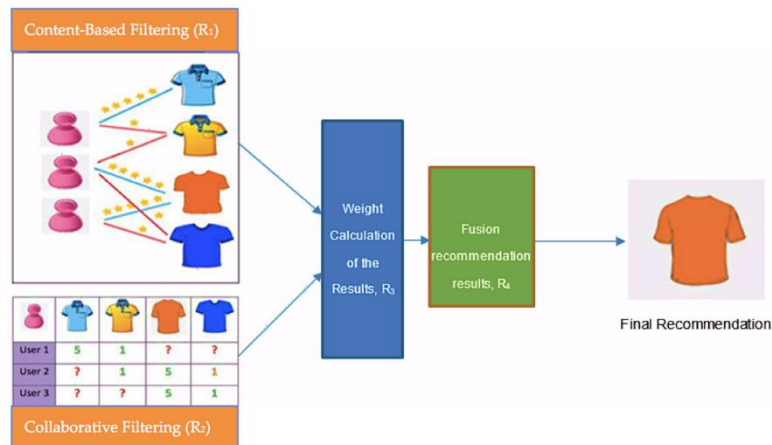
Thus, we will identifies patterns in user behavior to recommend items that similar users have liked.

Week 3: Hybrid Recommender System

In the third week, we will combine content-based and collaborative filtering methods to create a hybrid recommender system. This hybrid approach addresses the limitations of each individual method, providing more robust and personalized recommendations. By integrating both user-item interaction data and item metadata, the hybrid system can cater to a broader user base, including regular, active, inactive, premium, and non-premium members. For example, regular users with rich purchase histories will benefit from collaborative filtering, while inactive users or those with fewer interactions can still receive relevant recommendations through content-based filtering. This approach ensures that we cover a wide range of user behaviors and preferences.

I can use Weighted Hybrid where the scores from multiple recommendation methods (e.g., content-based and collaborative filtering) are combined using weighted averages to generate final recommendations.

I can also choose Ensemble methods like stacking or blending to combine the predictions from multiple models to improve overall recommendation accuracy and robustness.



Hybrid or Ensemble Model to give best performance

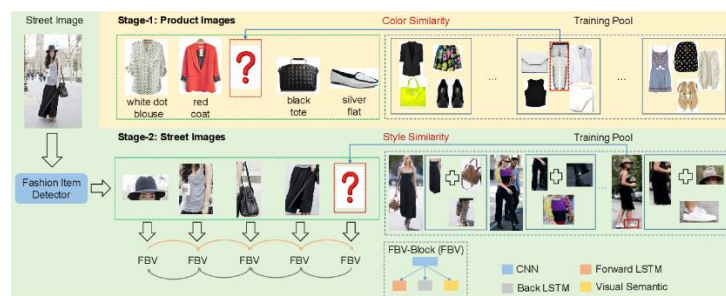
Week 4: Sequential and Context-Aware Recommendations

In the fourth week, we will explore sequential and context-aware recommendation techniques. Sequential recommendations consider the order of user interactions, capturing patterns in how users' preferences evolve over time. For instance, users might have different fashion needs in different seasons, and sequential models can adjust recommendations accordingly as per the temporal trends. Context-aware recommendations, on the other hand, incorporate additional contextual information such as location, time of day, and user activity. This is particularly relevant for fashion retail, as users' clothing preferences can vary based on these factors. Implementing these advanced techniques will further enhance the relevance and personalization of our recommendations. The H&M dataset has all these contextual information and I can take advantage of that.

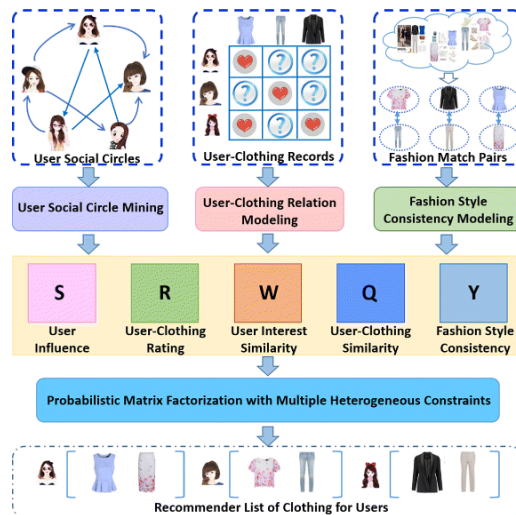
I can use methods like Convolutional Neural Networks and LSTM (Long Short Term Memory). They can capture the temporal dependencies in user behavior, making them suitable for predicting the next item a user might interact with based on their previous actions.

I can use Markov Chains etc. which model the probability of transitioning from one state (or item) to another. In the context of recommendations, they can predict the next item a user might interact with based on the current item. to implement these recommendation models.

I can also use Matrix Factorization for presenting the final context-aware recommendations.



Sequential trends can be recommended



Context Aware Recommendations

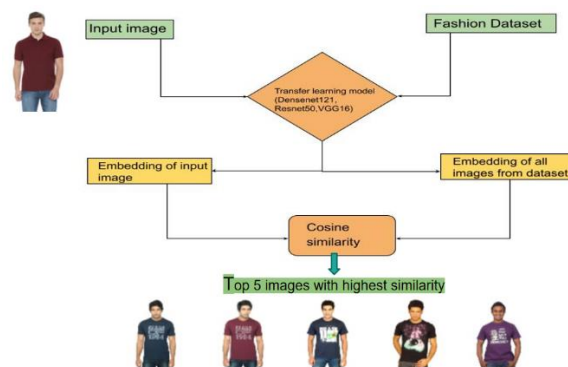
Week 5: Visual Similarity and Advanced Methods

In the fifth week, we will incorporate visual similarity techniques to enhance our recommendations. By using transfer learning models and cosine similarity, we can recommend items based on their visual features. This is particularly useful in the fashion domain where the appearance of items plays a crucial role in user preferences. This is **not a compulsory** step for the project. However, I am mentioning it as a draft proposal. This is not the same as Content Based Recommendation. This recommendation method will compare images of the fashion items themselves, as a final push to give fashion recommendation items to make the items more appealing to the customers.

I have an images folder with the dataset. This images folders has an image of each article i.e. fashion items. I can use the Similarity Methods like CNN's, kNN's or Transfer Learning methods like VGG16, ResNet or DenseNet which extract high-level visual features from fashion images. These features can be used to compute similarity scores between images of items. Thus, we get better recommendations.

Then, Neural Collaborative Filtering methods can be used can be extended to include contextual features. These models use embeddings to represent users, items, and context variables, capturing complex interactions and providing personalized recommendations based on the current context.

Finally, we can show the recommendations as their images of item to the user in a visual interface like mobile or website.



Visual Similarity methods to improve recommendations

Week 6: Evaluation Metrics and Refinement using User Feedback

In the final week, we will focus on evaluating the performance of the different methods and refining the models based on the results. We will use metrics such as Precision@K, Recall@K, Mean Average Precision (MAP), F1 Score etc. to assess the accuracy and relevance of the recommendations. Additionally, we will conduct A/B testing and do user studies to gather feedback on user satisfaction and engagement. This iterative process will help us fine-tune the recommendation models, ensuring they meet the needs of diverse user segments effectively.

4. Evaluation

To evaluate the performance of the fashion recommender system, we need to consider a variety of metrics that capture different aspects of recommendation quality. For the recommendation model, we will use metrics that focus on historical data, such as Precision@K, Recall@K, and Mean Average Precision (MAP).

Precision@N measures the proportion of recommended items in the top N recommendations that are relevant, while Recall@N measures the proportion of relevant items that are recommended in the top N results. MAP provides a single-figure metric to evaluate the overall ranking quality by averaging precision scores at different cut-off points. These metrics help us understand how well the model can recommend items that align with users' past interactions. These metrics are suitable for the H&M dataset because they allow us to measure the effectiveness of recommendations based on implicit feedback, such as purchase frequency and recency, which are prevalent in the dataset.

For the recommender system, since there will be real-time updation of user interactions, additional metrics can be used. Hit Rate and Success Rate will be used to evaluate how often the system correctly recommends items that users engage with. The Hit Rate measures the fraction of user sessions where at least one recommended item is interacted with, while the Success Rate considers the proportion of interactions that lead to successful outcomes, such as purchases or high ratings. These metrics can capture user engagement and satisfaction indirectly.

When evaluating the system, it is crucial to balance between different metrics, as focusing solely on one can lead to lower performance in others. For example, optimizing for precision might reduce recall, and vice versa. Therefore, we will consider the trade-offs between these metrics and use a combination of them to select the best model. In particular, we will compute F1-Score using the Precision@K and Recall@K metric values to provide a balanced view of the model's accuracy and coverage. MAP will also be important for understanding the overall ranking quality. This balanced approach ensures that the model performs well across various aspects, making it more robust and reliable.

Computational requirements are important, especially for real-time recommendations. The recommender system must be able to compute recommendations quickly to ensure a seamless user experience. Methods like matrix factorization and deep learning models require significant computational resources, so we will optimize these models for efficiency. We will use techniques like approximate nearest neighbors and model compression to reduce latency. This will be used to train the model fast in real time.

To evaluate the recommender system with real users, we will conduct a user study involving them. This study will involve presenting users with recommendations and collecting feedback through questionnaires. The feedback will focus on aspects such as satisfaction with the recommendations,

perceived relevance, and overall user experience. If we think about actually simulating a user interface, I can also gather data on user engagement metrics like CTR and Time Spent on Page during the study, assuming simulated user interfaces like mobile or website. Thus, using such user feedback, we can refine the recommendation model, ensuring that the system keeps on improving, and meets the needs of its users.